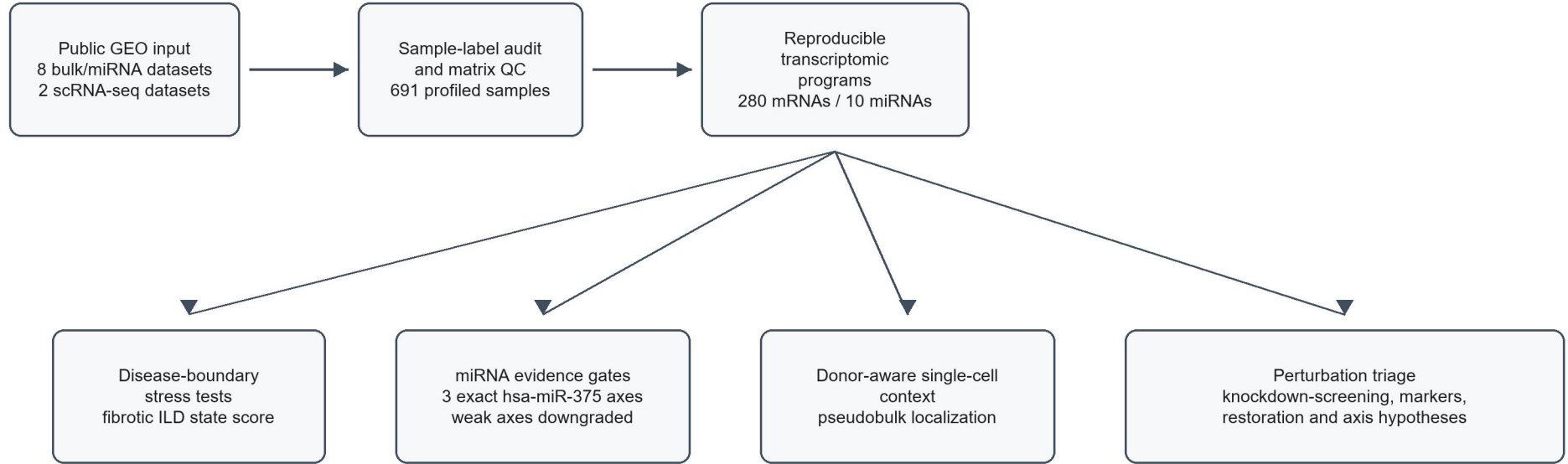


Boundary-tested perturbation-triage study design



All downstream gates use QC-passed datasets and preserve external validation cohorts for disease-state boundary testing.